

Introductions to Big Data Systems: Homework 12

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Introduction

The given problem is to compute a dilated convolution with halide. The definition of this operation is as following:

const int N = 5, CI = 128, CO = 128, W = 100, H = 80, KW = 3, KH = 3;

$$\text{dilated_conv}(n, h, w, oc) = \sum_{\{ic=0\}}^{\{IC-1\}} \sum_{\{kh=0\}}^{\{KH-1\}} \sum_{\{kw=0\}}^{\{KW-1\}} \text{input}(n, h + kh \cdot (dh + 1), w + kw \cdot (dw + 1), ic) \cdot \text{filter}(ic, kh, kw, oc)$$

Where in the input matrix is selected a $KW \cdot KH$ sub-matrix is selected, this selection is a square with dh and dw spaces between dots. Later with the filter matrix are multiplied over the input projection.

Implementation

vec

set to the hardware's natural width, ensuring efficient vectorized execution.

out

- **split(c, co, ci, vec):** splits the channel dimension so ci fits on vector.
- **reorder(ci, x, y, co, n):** puts the vectorized channel innermost for contiguous access.
- **vectorize(ci):** operates applies across channels.
- **parallel(n) and parallel(co):** parallelizes over batch and output-channel blocks for coarse-grain parallelism.

dilated_conv

- **compute_at(out, n):** computes the convolution per batch, improving cache locality and avoiding recomputation across batches.
- **vectorize(c):** Operates over channels during the reduction.
- **update:** Reduction schedule
- **reorder(r.x, r.y, r.z, c, x, y):** iterates spatial reduction first to maximize input reuse.
- **vectorize(c):** keeps channel in the reduction loop.
- **unroll(r.y) and unroll(r.z):** unrolls small kernel/dilation loops to reduce loops.
- **parallel(x) and parallel(y):** parallelizes over the input matrix by columns and row, there is not step dependance so there is not datarace in this steps.

Results

Dilation	Halide hw12	oneDNN
0	19.002888	281.723771
15	10.301415	274.332100
31	4.344090	257.487087
63	0.942404	231.179735

Table 1: Comparation in GFLOP/s between the student implementation and oneDNN